

Practice Exam Three: Version Two

Topics Covered:

- Properties of the Inductor and Capacitor
- RL/RC Circuits without Dependent Sources
- RL/RC Circuits with Dependent Sources

Practice Exam Tips:

- Treat this as timed practice, but do not rush. The goal is to identify gaps in understanding.
- After completing the exam, you can grade your work using the provided rubric in the solution. The practice solutions also include suggested discussion questions to try.
- If you make mistakes, revisit the relevant topic and rework the problem before reviewing the solution or attempting the second practice exam.
- Focus on method and clarity, not speed.

Exam Tips:

- Show all work clearly. Correct reasoning with minor algebra errors can still earn partial credit.
- Clearly label currents, voltages, and reference directions.
- Use consistent sign conventions throughout each problem.
- Include units in all final answers.
- The actual exam time limit is 75 minutes. If you follow the suggested pacing from this practice exam, you will finish in 60 minutes.

Question One: Properties of the Capacitor and Inductor

For the capacitive network shown below, connected to a black box, the following parameters are given:

$$v_b(0) = 8 \text{ V}$$

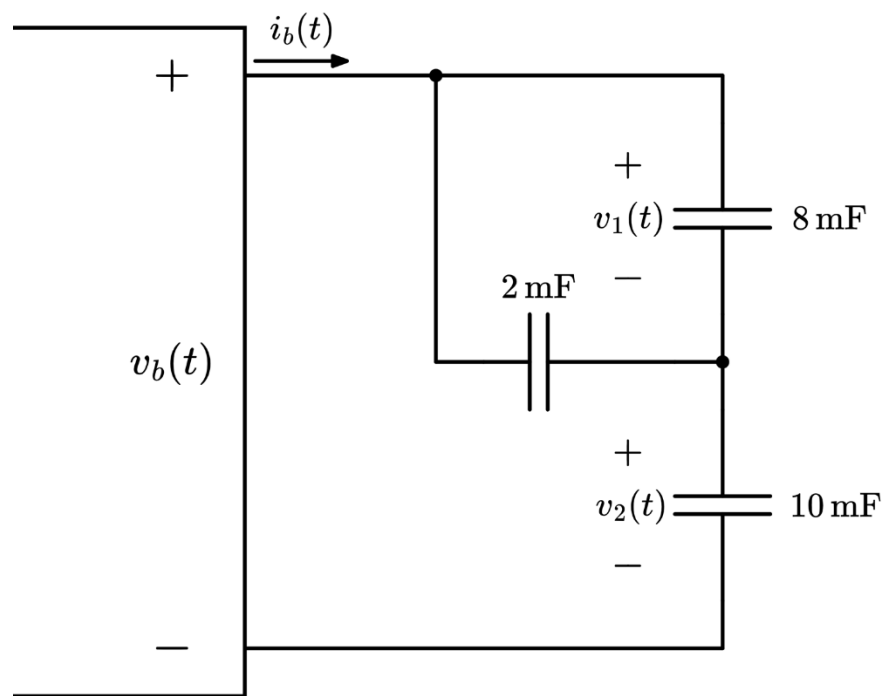
$$i_b(t) = 3e^{-3t} \text{ A}$$

$$v_1(0) = 5 \text{ V}$$

Calculate the following expressions, seen by the black box:

- a) The equivalent capacitance
- b) $v_1(t)$
- c) $v_2(t)$

Suggested time limit: 20 minutes

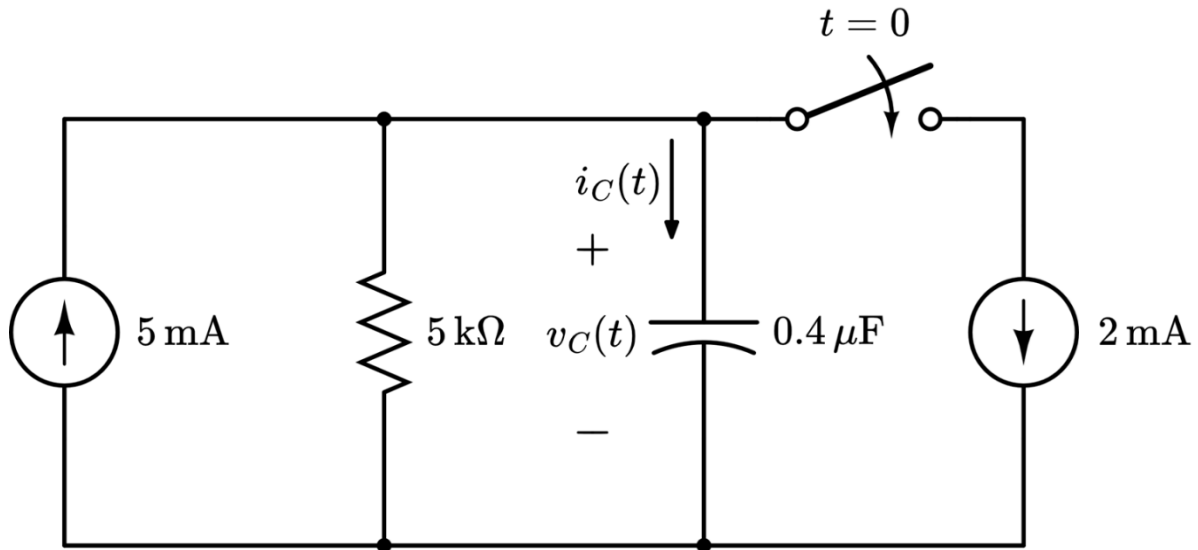


Question Two: RL/RC Circuits without Dependent Sources

For the circuit shown below, the switch has been open for a long time and closes at $t = 0$. Find the following expressions:

- a) $v_C(0)$
- b) $v_C(t)$
- c) $i_C(t)$

Suggested time limit: 20 minutes



Question Three: RL/RC Circuits with Dependent Sources

For the circuit shown below, the switch has been open for a long time and closes at $t = 0$. Find the following expressions:

- The initial current $i_L(0)$.
- The steady-state expression for the inductor current $i_L(t)$ for $t \geq 0^+$.
- The steady-state expression for the resistor current $i_o(t)$ for $t \geq 0^+$.

Suggested time limit: 20 minutes

